📝 Worksheet W13: Mixed Review

"Comprehensive Practice — Test Your Knowledge Across All Chapters"

All Chapters · 15 Problems · Name: ______________________ Date: ______________________

📊 Topics Covered: Functions · Limits · Slope · Power Rule · Integrals · Area · Fundamental Theorem · e^x · Sine/Cosine · Optimization

Problem M1

Question: What are the **two questions** that all of calculus answers?

💡 *Hint: Think about speedometer and odometer.*

Problem M2

In the function $y = 3x + 2$, what is y when $x = 4$?

💡 *Hint: Multiply x by 3, then add 2.*

Problem M3

For $d = t^2$, what is the **instant speed** at $t = 4$ seconds?

💡 *Hint: The derivative of t^2 is $2t$.*

Problem M4

Find the slope between the points $(1, 5)$ and $(4, 14)$.

Hint: Slope = Rise ÷ Run.

Problem M5

Find the derivative of $f(x) = x^5$.

Hint: Bring the exponent down in front, then subtract 1.

Problem M6

Find the derivative of $f(x) = 3x^4$.

Hint: Coefficient × exponent, then reduce exponent by 1.

Problem M7

Find the antiderivative of $2x$.

Hint: What function has derivative $2x$? Don't forget $+C$.

Problem M8

Find the area under $y = x^2$ from $x = 0$ to $x = 3$ using the Fundamental Theorem.

Hint: Antiderivative of x^2 is $x^3/3$.

Problem M9

Question: What is the derivative of e^x?

Hint: e^x is its own derivative.

Problem M10

Find the derivative of $f(x) = e^{3x}$.

Hint: The derivative of e^{kx} is $k \times e^{kx}$. Here $k = 3$.

Problem M11

Question: What is the derivative of $\sin(x)$?

Hint: The derivative of sine is cosine.

Problem M12

Find the derivative of $f(x) = \cos(2x)$.

Hint: derivative of $\cos(kx)$ is $-k\sin(kx)$. Here $k = 2$.

Problem M13

For $f(x) = -x^2 + 6x$, find the critical point.

Hint: Find $f'(x)$ and set it to zero.

Problem M14

A rectangle has a perimeter of 40 meters. What is the **maximum possible area**?

Hint: The square maximizes area.

Problem M15

Question: Explain the **Fundamental Theorem of Calculus** in your own words. Why is it called the "Great Reversal"?

Hint: Before the theorem, slope and area were seen as separate. What did the theorem reveal?